

SKIN DISEASE CAUSES & TREATMENT

Actinic keratosis

What is Actinic keratosis?

Actinic keratosis also known as solar keratosis, is a skin condition that happens with too much sun exposure. It causes rough, scaly spots of skin. They commonly appear on your:

- Scalp
- Face (lips, nose, ears)
- Arms
- The back of your hands

But it can form on other sun-exposed areas, too.

It's the most common type of skin precancer. It can lead to a type of skin cancer called [squamous cell carcinoma](#). Treatment is available to prevent actinic keratosis from turning into skin cancer.

Symptoms :

Signs and symptoms of actinic keratosis

If you have this skin condition, you might notice an area of skin that's:

- Bleeding
- Bumpy
- Burning, stinging or uncomfortable under clothing
- Chapped (lower lip)
- Discolored (darker than your skin tone, red, pink, brown or gray)
- Itchy, tender or painful to the touch
- Rough, dry, scaly or crusty

Actinic keratosis causes:

Too much [exposure to ultraviolet \(UV\) light](#) causes actinic keratosis. It's most often from the sun. It can also develop from frequent use of indoor tanning equipment, like tanning beds.

UV rays damage the DNA in your skin cells. It targets the outer layer of cells called keratinocytes. Over time, repeated sun exposure causes these cells to grow abnormally. This leads to a rough or scaly spot on your skin.

Risk factors of actinic keratosis:

You may be more likely to get this skin condition if you:

- Are [male](#)
- Have a [compromised immune system](#)
- Have a light skin tone
- Have a history of actinic keratosis or skin cancers
- Have had many years of sun exposure or tanning bed use
- Live closer to the equator

Actinic keratosis treatment:

Treatment options depend on how many actinic keratoses you have and where they are on your body. Options include:

- [Laser therapy](#): This uses focused light to destroy damaged skin cells. It may help with multiple AKs or larger areas.
- [Cryotherapy](#): This freezes off the spot using liquid nitrogen. It works best if you have just a few AKs. The area may blister and peel off after a few days.
- [Photodynamic therapy](#): This uses a light-sensitive cream and special light to target AKs. It's often used for multiple or recurring spots. You'll need to avoid sunlight for a few days while your skin heals.
- [Chemical peel](#): This removes damaged skin with a special chemical solution. The top layer of skin peels off, and new, healthy skin forms as it heals.
- **Curettage**: A provider scrapes or burns off the damaged area. Your provider may use this with heat or cryotherapy to fully remove the spot.
- [Surgical excision](#): Your provider cuts out the spot. They'll numb the area first and may need to use stitches. Your skin usually heals in two to three weeks.
- using removal treatments and medicated creams.

Prevention:

The best way to prevent actinic keratosis is to limit your exposure to sunlight. You can protect your skin by:

- Avoiding tanning beds or indoor UV lamps
- Staying out of the sun between 10 a.m. and 4 p.m., when UV rays are the strongest
- Wearing protective clothing, like long-sleeved shirts, long pants, wide-brimmed hats and sunglasses
- Wearing sunscreen with SPF 30 or above every day, even when it's cloudy or cold

Basal cell carcinoma

What is Basal Cell Carcinoma?

Basal cell carcinoma (BCC) is a type of [skin cancer](#) that forms in the basal cells of your skin. Basal cells exist in the lower part of your epidermis, which is the outside layer of your skin. Basal cell carcinoma looks like a small, sometimes shiny bump or scaly flat patch on your skin that slowly grows over time.

Symptoms and Causes

Signs of basal cell carcinoma include:

- Lumps, bumps, pimples, scabs or scaly lesions on your skin
- The lump may be slightly see-through (translucent) and close to your normal skin color or white to pink, brown to black or black to blue.
- The lump may appear shinier than the skin around it with tiny visible blood vessels.
- The lump may grow slowly over time.
- The lump may be itchy or painful.
- The lump may form an ulcer, which can ooze clear fluid or bleed with contact.

The most common places to have BCC include:

- Face
- Scalp
- Nose
- Eyelids
- Legs
- Ears
- Arms.

Causes:

A change to your DNA causes basal cell carcinoma. This change usually happens after your skin has [too much exposure to ultraviolet \(UV\) rays](#) from sunlight or tanning beds.

This causes your basal cells to make too many cells, which causes lumps (tumors) or lesions to form in the outer layer of your skin (epidermis).

A rare inherited condition called basal cell nevus syndrome ([Gorlin syndrome](#)) causes BCC to appear in childhood.

Treatment:

- **Electrodesiccation and curettage:** Scraping off the cancerous lump with a curette and then burning with a special electric needle.
- **Surgery:** Removing the cancerous lump or lesion with a scalpel (excision or [Mohs surgery](#)).
- **Cryotherapy or cryosurgery:** Freezing the cancerous lump to remove it.
- **Chemotherapy:** Using powerful medicines to kill cancerous cells in your body.
- **Photodynamic therapy (PDT):** Applying blue light and a light-sensitive

What happens if basal cell carcinoma is left untreated?

If you don't receive treatment for basal cell carcinoma, the skin cancer can slowly grow in size and invade deeper tissues like muscle and bone and cartilage. The BCC may become painful and ulcerated, which can cause bleeding and infection.

In extremely rare cases, basal cell carcinoma can spread to other parts of your body and cause life-threatening side effects.

Prevention:

While all cases of basal cell carcinoma can't be prevented, you can take steps to reduce your risk by:

- Avoiding sun exposure from 10 a.m. to 4 p.m.
- Avoiding tanning beds.
- Using sunscreen with an SPF of 30 or higher each day and reapply sunscreen every two hours throughout the day if you're outdoors or participating in activities like swimming.
- Wearing clothing that has built-in sun protection (UPF), broad-brimmed hats and sunglasses.
- Performing a skin self-exam once per month to check for any unusual lumps or lesions.
- Visiting a [dermatologist](#) annually for a skin examination.
- Contacting your healthcare provider if you have any questions about your skin or changes to your skin.
- Taking nicotinamide (vitamin B3) 500 milligrams twice a day can reduce the risk of developing new BCC and SCC.

Dermatofibroma

What is dermatofibroma?

Cellular dermatofibromas are a type of dermatofibroma (noncancerous tumor). They appear as firm bumps (nodules), often on your legs.

dermatofibromas are superficial, meaning they grow on the top layer of skin ([epidermis](#)).

Compared to other dermatofibromas, cellular dermatofibromas are more likely to come back (recur) after treatment. In about 1 in 10 people, cellular dermatofibromas cause necrosis in the tumor. Necrosis is tissue or cell death in a certain area of your body.

Dermatofibromas are most common in people ages 20 to 49. Twice as many women as men have dermatofibromas.

Symptoms and Causes

symptoms of dermatofibroma:

Cellular dermatofibromas look like small round or oval bumps. They're usually less than 1 centimeter (cm) in diameter. They may be flesh-colored, brown or reddish-brown.

Usually, dermatofibromas form on your arms or legs. They may feel firm to the touch and can be itchy, tender or painful. Sometimes, they're shaped like a dimple — the center of the lesion falls in, and the edges of the lesion pucker around it.

Causes of dermatofibroma:

People with conditions that affect their [immune system](#) may have multiple dermatofibromas in a cluster. You're more likely to have a dermatofibroma cluster if you have:

- HIV/AIDS
- Leukemia
- Lupus

Treatment:

Because they're noncancerous, dermatofibromas don't always require treatment.

If a dermatofibroma is large or causes discomfort, your healthcare provider may remove it. Removal is a short in-office procedure. They may use:

- [Steroid injections](#) to reduce pain or lesion size.
- Surgical excision, using a surgical tool to scrape off the lesion.

Prevention:

There isn't a known way to prevent a cellular dermatofibroma. Fortunately, dermatofibromas aren't cancerous. If you get one, it rarely causes discomfort or harmful symptoms.

Melanoma

What is melanoma?

Melanoma, which means "black tumor," is the most dangerous type of [skin cancer](#). It grows quickly and has the ability to spread to any organ.

Most melanomas are black or brown in color, but some are pink, red, purple or skin-colored. About 30% of melanomas begin in existing [moles](#), but the rest start in normal skin.

how many moles you have may help predict your skin's risk for developing melanoma. It's important to know if you're in a high-risk group for developing melanoma skin cancer. Because of the fast growth rate of melanomas, a treatment delay sometimes may mean the difference between life and death.

Knowing your risk can help you be extra vigilant in watching changes in your skin and seeking skin examinations since melanomas have a 99% cure rate if caught in the earliest stages.

Symptoms and Causes

signs of melanoma:

- [Melanoma can appear as moles](#)
- scaly patches

- open sores
- raised bumps

Causes of melanoma:

Most experts agree that a major risk factor for melanoma is [overexposure to sunlight](#), especially sunburns when you are young.

UV radiation from tanning beds also increases the risk of melanoma and has been designated as a carcinogen (cancer-causing) by the World Health Organization. An increased risk for developing the disease is seen in people with:

- Fair skin, freckles, blond or red hair and blue eyes.
- Excess sun exposure, including blistering sunburns.
- An address near the equator or in high elevations — living in these locations may increase your UV exposure.
- A history of tanning bed use.
- A family history of melanoma.
- Many moles, especially [atypical moles](#).
- A weakened immune system.

Treatment:

Your melanoma treatment will depend on the stage of the melanoma and your general health.

- **Melanoma Surgery:** In the early stages, surgery has a high probability of being able to cure your melanoma. Usually performed in an office, a dermatologist numbs the skin with a local anesthetic and removes the melanoma and margins (healthy surrounding skin).
- **Lymphadenectomy:** In cases where melanoma has spread, removal of the lymph nodes near the primary diagnosis site may be required. This can prevent the spread to other areas of your body.
- **Metastasectomy:** Metastasectomy is used to remove small melanoma bits from organs.
- **Targeted cancer therapy:** In this treatment option, drugs are used to attack specific cancer cells. This “targeted” approach goes after cancer cells, leaving healthy cells untouched.
- **Radiation Therapy:** Radiation therapy includes treatments with high-energy rays to attack cancer cells and shrink tumors.
- **Immunotherapy:** immunotherapy stimulates your own immune system to help fight the cancer.

Prevention:

You may reduce your risk of melanoma by protecting yourself from excess sun and sunburns.

- Avoid sun and seek shade, especially between 10 a.m. and 4 p.m.
- Don't use tanning beds. Use a spray tan (cosmetic) instead.
- Whenever possible, wear hats with brims, sunglasses, long-sleeved shirts and pants.
- Use a broad-spectrum sunscreen with a skin protection factor (SPF) of 30 or higher and reapply often, usually every 1.5 hours or more often if you're swimming or sweating.
- Use a lip balm with sunscreen.
- Don't forget to apply sunscreen to young children and infants older than 6 months.

Healthy foods to consider:

- **Daily tea drinking:** The polyphenols (antioxidants found in plants) in tea help strengthen your immune system. Green tea contains more polyphenols than black tea.
- **High vegetable consumption:** Eating carrots, cruciferous and leafy vegetables is linked to the prevention of cutaneous (invasive) melanoma.
- **Weekly fish intake:** Study participants who ate fish weekly seemed to avoid developing the disease when compared to those who did not eat fish weekly.

Nevus

What is a nevus?

Most people have between 10 and 40. Common nevi are harmless collections of colored cells. They typically appear as small brown, tan, or pink spots.

Some of them are harmless and others more serious. Sun exposure has also been linked to the formation of some types of nevi.

Symptoms and Causes

Here are some of the most common signs and symptoms of a nevus:

- Darkening of skin in a specific area, usually between 1 and 5 cm across
- Patches of extra skin growth over time
- Reddened patches on the skin
- Raised bumps on the skin surface
- Itching around the affected area
- Painful feeling in certain areas when touched

Causes:

- Most common is genetic
- Repeated trauma or injury to the area
- Immunological disorders such as lupus
- Environmental factors like pollution or toxins
- In some cases, the cause may be unknown.

Treatment:

- **Observation:** This is the most common approach and involves regular check-ups with your doctor to monitor the size and shape of the nevus. It may become darker in color or more raised over time; any changes should be reported immediately to your doctor for further evaluation.
- **Surgical Excision:** Surgical excision involves removing the entire growth using a scalpel or other surgical tools. Depending on the size and location, it can be done under local or general anesthesia. After excision, the nevus is typically sent to a laboratory for evaluation.
- **Laser Therapy:** Laser therapy involves using targeted energy in the form of light to reduce the size of a nevus. It is usually used for very small, lighter-colored nevi that are not located deeper in the skin's layers.
- there are also some over-the-counter creams and lotions available that may help lighten dark spots caused by nevi.

Talk to your doctor, dermatologist, or plastic surgeon to decide which treatment is right for you.

Prevention:

- **Avoid Smoking:** Smoking can slow down the healing process, increase complications and the risk of cancer.
- **Avoid Sun Damage:** Wear sunscreen with an SPF of at least 30 and limit your exposure to direct sunlight as much as possible.
- **Get Plenty of Rest:** Sleep is essential for the body to heal from any injury or condition, so make sure you get enough rest each night.
- **Exercise Regularly:** Exercise can help improve circulation in the affected area as well as reduce stress, which can aggravate a nevus.
- **Eat Healthy:** It promotes healing, boosts immune function and helps to manage pain and inflammation.

Pigmented benign keratosis

What is seborrheic keratosis?

Pigmented benign keratosis often called seborrheic keratosis is a common benign skin growth, similar to a mole. Most people will have at least one in their lifetime. They tend to appear in mid-adulthood and their frequency increases with age.

They are harmless and don't require treatment, but you can have them removed if they bother you.

Symptoms and Causes

They usually don't, but people sometimes report:

- Itching.
- Irritation from friction.
- Bleeding

If these symptoms annoy you, you may want to have the growth removed.

Treatment:

If it is clearly a seborrheic keratosis, it won't require any treatment. But you might want to have it removed several options are:

- **Electrodesiccation/Curettage:** Your healthcare provider will numb the skin and then use a targeted electrocurrent to burn the seborrheic keratosis.
- **Laser Therapy.** Lasers offer an alternative to surgery by burning the growth, sterilizing the wound and sealing the tissue all at once. Laser therapy is quick, but the wound will be sore for a while afterward.
- **Prescription Hydrogen Peroxide.** The FDA has recently approved a topical solution of 40% hydrogen peroxide to treat seborrheic keratosis. You may need more than one visit to see results.
- **Cryotherapy.** Your healthcare provider will numb the skin and then use liquid nitrogen to freeze the growth. This will cause it to fall off within a few days or weeks. One possible side effect is that the skin where the growth was will lose some of its pigment and look lighter

Squamous cell carcinoma

What is Squamous Cell Carcinoma?

Squamous cell carcinoma (SCC) or cutaneous squamous cell carcinoma (CSCC) is the second most common form of [skin cancer](#) after basal cell carcinoma. Usually, squamous cell carcinomas form on areas of your skin that receive the most sun exposure like your head, arms and legs.

Cancer can also form in areas of your body where you have mucous membranes, which are the inner lining of your organs and body cavities like in your mouth, lungs and anus.

Risk Factors of Squamous cell carcinoma:

Squamous cell carcinoma can affect anyone. You're most at risk if you:

- Have long-term [sun exposure](#) or sun damage to your skin at a young age.
- Have a pale complexion, blue or green eyes, blonde or red hair.
- Are 65 years of age or older.
- Have a weak immune system or received an organ transplant.
- Had chemical exposure (cigarettes, arsenic).

Symptoms and Causes

Symptoms of squamous cell carcinoma include skin changes like:

- A rough-feeling, bump or growth, which might crust over like a scab and bleed.
- A growth that's higher than the skin around it but sinks down (depression) in the middle.
- A wound or sore that won't heal, or a sore that heals and then comes back.
- An area of skin that's flat, scaly and red that's larger, about 1 inch (2.5 centimeters).
- White or pale spots in your mouth, on your tongue, gums or cheeks
- A lesion on your lower lip where the tissue becomes pale, dry and cracked

Causes:

A mutation to the *p53* gene causes squamous cell carcinoma. The most common way that your *p53* gene mutates is from ultraviolet (UV) exposure from the sun, or from using indoor tanning beds.

Treatment:

Treatment for squamous cell carcinoma focuses on removing cancer from your body.

- **Cryosurgery**: Freezing the cancer cells to destroy them.
- **Photodynamic therapy (PDT)**: Using blue light and light-sensitive agents to remove cancer from your skin.
- **Curettage and electrodesiccation**: Scratching off the cancerous lump with a spoon-like instrument (curette), then burning the area with an electric needle.
- **Excision**: Cutting the cancer out of your skin and stitching your skin back together.
- **Mohs surgery**: Removing layers of skin affected by cancer, most common for facial cancers.
- **Systemic chemotherapy**: Using powerful medicines to destroy cancer cells in your body.

Prevention:

While you can't prevent all types of squamous cell carcinoma, you can take steps to reduce your risk by:

- Avoiding excessive sun exposure.
- Avoid using tanning beds.
- Using sunscreen when you're outdoors.
- Wearing protective clothing and accessories from the sun, including sunglasses, hats and clothing with UPF protection.
- Stopping smoking.
- Avoid exposing yourself to chemicals without wearing personal protective equipment.

Vascular lesion

What is vascular lesion?

Vascular lesions are abnormal clusters of blood vessels (hemangiomas) or malformations (capillary, venous, lymphatic) appearing as red/purple skin marks, lumps, or deep tissue issues.

Common Symptoms:

- **Skin Discoloration:** Red, purple, or pink marks (e.g., port-wine stains).
- **Visible Lumps:** Soft, raised masses underneath the skin.
- **Pain and swelling:** Common with venous and arteriovenous malformations.
- **Functional Issues:** Breathing, vision, or feeding difficulties if located near the eyes, mouth, or airway.
- **Bleeding/Ulceration:** Lesions may bleed or form sores.

Causes:

Most vascular malformations are a result of localized or regional abnormal development of vascular and/or lymphatic channels which develop in utero.

Less commonly, an injury in which there's unrecognized damage to the vessels can over time develop into a vascular malformation.

Risk Factors of vascular lesion:

Some people inherit gene changes that cause conditions that increase the risk of vascular malformations. These conditions include:

- [Hereditary hemorrhagic telangiectasia](#) (also called Osler-Weber-Rendu syndrome) causes abnormal tangles of small capillaries or AVMs.
- [Klippel-Trenaunay syndrome](#) (KTS) causes port-wine stains, bony or tissue growths and venous or lymphatic malformations
- [Parkes Weber syndrome](#) causes AVMs in your arms or legs.
- **Blue rubber bleb nevus syndrome** causes venous malformations in your intestines and [digestive system](#).

Treatment:

Treatments depend on the affected blood vessels. They often involve closing off or surgically removing affected blood vessels.

- [Sclerotherapy](#).
- Catheter embolization.
- Laser treatments.
- [Radiation therapy](#) (radiosurgery).

Sometimes, your healthcare provider may recommend a biopsy to confirm the diagnosis and/or to obtain tissue for genetic testing.

Prevention:

Vascular malformations are often present at birth (congenital). That means there isn't a way to prevent them.

If you have a family history of a condition that causes vascular malformations, you may want to meet with a genetic counselor. This specialist can discuss options to lower the risk of passing the condition to children.